Section IV.A: non-Abelian fractons from gauging transversal CZ gates

The SWAP example established the graph-gauging mechanism in its most universal form. This example is more directly connected to quantum computation: it gauges a transversal CZ, the logical Clifford operator whose measurement will later prepare a magic state. It also lets us follow the construction on a concrete three-dimensional lattice.

There are two things to keep distinct from the outset. The **ALP code** is an Abelian CSS code with fracton/subsystem structure. The construction then takes two copies of that code and gauges a symmetry between the copies. The gauged code is the non-Abelian object. In this first pass we will follow the graph construction through its local circuit and its resulting constraints. The later subsections on addressable gates and cup products are deliberately postponed: they answer how to gauge a family of CZ gates, not what the basic CZ-gauging mechanism is.

The unusual qubit placement in the ALP code

The anisotropic lineon-planeon (ALP) model lives on a periodic cubic lattice. Its CSS chain complex is

$$0 \longrightarrow D_2 \xrightarrow{\partial_2} D_1 \xrightarrow{\partial_1} D_0 \longrightarrow 0. \tag{169}$$

The word *anisotropic* is literal: the three spatial directions are not treated alike. The bases have the following geometric interpretation:

space	basis object	role
D_2	cubes c	Z-check labels
D_1	plaquettes in xy planes and edges parallel to z	physical qubits
D_0	vertices v	X-check labels.

(170)

The middle line is the surprise. A horizontal *xy plaquette* carries a qubit, whereas a vertical *z edge* carries a qubit. There are no qubits on horizontal edges. This is why the ALP stabilizers do not look like the usual cubic-lattice toric-code stabilizers.

The drawings below are reproduced directly from the paper's TikZ source, with the geometry unchanged. The left drawing gives a cube Z-check; the right drawing gives a vertex X-check. A letter marks a qubit on which that Pauli acts. In particular, a letter inside a horizontal face means a **face qubit**, not a qubit at a hidden vertex.

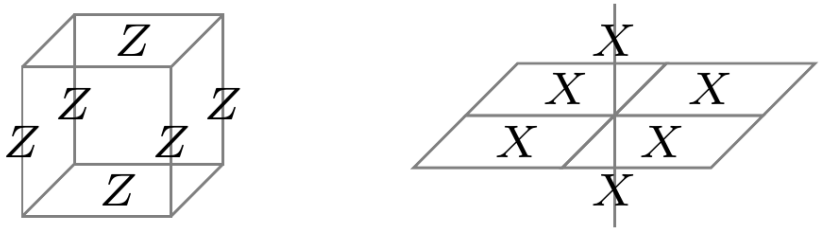


Figure 1: ALP cube Z check (left) and vertex X check (right), reproduced from the paper.

Source: Section IV.A, equation labelled `eq:qubit-ALP-stabilizers` *in the paper's source. Each check acts on six qubits.*

To make these perspective drawings unambiguous, label a vertex by integer coordinates (i, j, k). Write $p_{i,j,k}$ for the horizontal square with corners $(i, j, k), (i+1, j, k), (i, j+1, k), (i+1, j+1, k)$, and $\ell_{i,j,k}$ for the vertical edge from (i, j, k) to $(i, j, k+1)$. Coordinates are understood periodically. The cube $c_{i,j,k}$ then has the following check:

$$B_{i,j,k} = Z_{p_{i,j,k}} Z_{p_{i,j,k+1}} \prod_{\alpha,\beta\in\{0,1\}} Z_{\ell_{i+\alpha,j+\beta,k}}.$$

The two face factors sit on the bottom and top; the other four factors sit on the upright edges. At a vertex, the check is instead

$$A_{i,j,k} = X_{\ell_{i,j,k-1}} X_{\ell_{i,j,k}} \prod_{\alpha,\beta\in\{0,1\}} X_{p_{i-\alpha,j-\beta,k}}.$$

Here the two edge factors are below and above the vertex, and the four horizontal squares meet at that vertex. These formulas specify every qubit in the drawings without requiring the reader to reconstruct depth from a flat image.

The chain maps encode exactly these incidences. The map ∂_2 sends a cube to the sum of its two horizontal faces and four vertical edges. The map ∂_1 sends a face qubit to its four corner vertices, and an edge qubit to its two endpoints. This is a **code chain complex**: its degree records a qubit's role in the checks, not necessarily the spatial dimension of the object carrying it. Both kinds of qubit belong to D_1, even though a square is geometrically two-dimensional.

For example, applying $\partial_1\partial_2$ to a cube counts each of its eight corners twice: once from a horizontal face, once from a vertical edge. Over $\mathbb{F}_2$, these contributions cancel. Equivalently, an incident vertex check and that cube check share two qubits, so commuting them produces $(-1)^2 = +1$. This is the concrete content of the chain-complex condition in this model.

In the paper's compact notation, the two stabilizer families are therefore

$$A_v = X_{d_0 v}, \qquad B_c = Z_{\partial_2[c]}. \tag{171}$$

The chain-complex condition $\partial_1\partial_2 = 0$, or dually $d_1 d_0 = 0$, guarantees that each A_v and B_c overlap on an even number of qubits and hence commute. This establishes that the checks define a CSS code. Gauging will additionally require a symmetry preserving that code, which we verify below.

An excitation is a check with eigenvalue -1, rather than the ground-state value $+1$. Restricted mobility means that transporting an isolated excitation in some directions requires creating additional excitations. A *lineon* can move along a line; a *planeon* can move within a plane. For a concrete first example, $Z_{\ell_{i,j,k}}$ anticommutes with the vertex checks at the two ends of that edge. A product of Z's along a vertical segment leaves defects only at its endpoints, because the interior signs flip twice. In contrast, $Z_{p_{i,j,k}}$ creates four vertex defects at that square's corners. This already shows why vertical and horizontal operations have different effects. The broader term *fracton order* includes systems with such restricted-mobility excitations; it does not mean that every excitation is completely immobile.

Here each check has weight six, and each qubit occurs in a bounded number of checks, independently of lattice size. That is the low-density condition in qLDPC. "Subsystem" in the discussion of spatial symmetries should not be confused with a subsystem error-correcting code with noncommuting gauge checks: the starting ALP code is a commuting CSS stabilizer code.

For an $L_x \times L_y \times L_z$ periodic lattice, the paper reports $2(L_x + L_y - 1)$ logical qubits and distance scaling as $\min(L_x, L_y, L_z)$. This is a useful reminder that it is a subsystem-like 3D code rather than a conventional topological code with a constant number of logical qubits.

The transversal CZ symmetry between two copies

The next step is to identify a gate that preserves the two input code spaces. This matters because gauging an arbitrary collection of physical gates would not automatically produce a logical-symmetry measurement.

Make two identical ALP blocks, named B and C. In the paper's original drawing, blue dots belong to B, orange dots to C, and black segments specify the CZ pairs. Repeat the displayed pattern in every cubic cell.

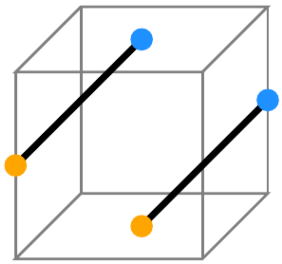


Figure 2: Original ALP transversal CZ pairing from Section IV.A of the paper.

Source: the paper's equation labelled `eq:cubic-CZ-sym`*. Colours distinguish code blocks, not face qubits from edge qubits.*

Writing $\mathcal{P}$ for this matching gives

$$U_{\mathsf{CZ}} = \prod_{(e,e')\in\mathcal{P}} \mathsf{CZ}_{B_e,C_{e'}}. \tag{172}$$

In this transversal matching, each physical qubit participates in exactly one two-qubit gate between blocks. Merely having a bounded number of gates per qubit would describe a bounded-depth circuit, a weaker condition. The paired qubits need not be drawn directly on top of one another: the pairing is a local matching determined by the cubic unit cell. This is a point at which the paper's 3D perspective matters more than an attempted planar picture.

The elementary conjugation rules are

$$\mathsf{CZ}_{B,C} X_B \mathsf{CZ}_{B,C} = X_B Z_C, \qquad \mathsf{CZ}_{B,C} X_C \mathsf{CZ}_{B,C} = Z_B X_C, \tag{173}$$

while both Z_B and Z_C are unchanged. The chosen pairing makes the extra Z factors fit the ALP check structure. The code-preserving property concerns the complete gate: a *single physical* X_e, Z_e, or $\mathsf{CZ}_{B_e,C_{e'}}$ is generally **not** a symmetry of the code. For example, a lone Z_e anticommutes with the X-checks incident on that qubit and creates syndrome defects. Equation (173) is only a local operator identity; it does not say that its individual factors preserve check eigenvalues.

The coordinate convention above makes it possible to check the symmetry explicitly. In cell (i, j, k), the two pairs in the drawing are

$$(C_{p_{i,j,k}}, B_{\ell_{i+1,j+1,k}}), \qquad (C_{\ell_{i,j,k}}, B_{p_{i,j,k+1}}).$$

Consider the entire B-vertex check at (i, j, k). Its two vertical edge qubits acquire C-block Z's on the horizontal faces $p_{i-1,j-1,k-1}$ and $p_{i-1,j-1,k}$. Its four face qubits acquire Z's on the four vertical edges of the same cube. The six new factors are therefore **exactly one complete cube check**:

$$\begin{aligned} U_{\mathsf{CZ}} A^B_{i,j,k} U^\dagger_{\mathsf{CZ}} &= A^B_{i,j,k} B^C_{i-1,j-1,k-1}, \\ U_{\mathsf{CZ}} A^C_{i,j,k} U^\dagger_{\mathsf{CZ}} &= B^B_{i,j,k} A^C_{i,j,k}. \end{aligned} \tag{174}$$

For the second identity, the two C-edge qubits attach Z's to the bottom and top B-faces of $c_{i,j,k}$; the four C-face qubits attach them to its four upright B-edges. This verifies both directions of the matching. The mechanism is the assembly of complete checks, rather than a claim that each physical X or each elementary gate preserves the code separately. For products of checks, repeated Pauli factors cancel modulo two in the usual way.

On a code state, both factors on either right-hand side have eigenvalue $+1$. Thus the transformed check still fixes the state: the transversal gate maps the stabilizer group to itself. It can change the encoded logical state while keeping the physical state inside the codespace.

The Z-checks themselves do not acquire anything, because CZ is diagonal:

$$U_{\mathsf{CZ}} B^B_c U^\dagger_{\mathsf{CZ}} = B^B_c, \qquad U_{\mathsf{CZ}} B^C_c U^\dagger_{\mathsf{CZ}} = B^C_c. \tag{175}$$

In binary language, the test for a proposed matching is that its added Z support be in $\operatorname{im} \partial_2$ for every X-check support in $\operatorname{im} d_0$.

The cubic ancilla graph and the controlled-CZ enrichment

Having verified the matter symmetry, we now introduce the ancillary system that will turn its global action into local gauge constraints. Put temporary qubits a_v on every cubic-lattice vertex and gauge-field qubits A_f on every cubic-lattice edge $f = \langle v, w \rangle$. After making the graph cluster state (the paper applies a Hadamard to the A sublattice), its convenient stabilizer generators are

$$K^a_v = X^a_v \prod_{f \ni v} X^A_f, \qquad K^A_f = Z^A_f Z^a_v Z^a_w. \tag{176}$$

Because the cubic graph is connected, the temporary vertices possess exactly one global X symmetry,

$$X^a_{\text{all}} = \prod_v X^a_v. \tag{177}$$

The enrichment must tie this single symmetry to Eq. (172). It uses CCZ, the three-qubit controlled-controlled-Z gate. Its action is diagonal in the computational basis:

$$\mathsf{CCZ}_{a,B,C} |r, b, c\rangle = (-1)^{rbc} |r, b, c\rangle. \tag{178}$$

Thus when the temporary control is $|0\rangle_a$, it does nothing; when it is $|1\rangle_a$, it applies CZ to the selected B, C pair. The paper's original enrichment drawing shows exactly which two pairs are controlled by the purple vertex:

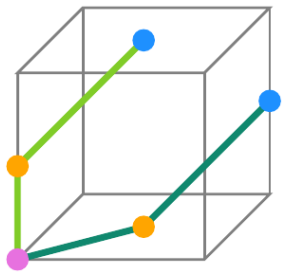


Figure 3: Original ALP CCZ enrichment: two green paths connect the purple control to the two matter pairs.

Source: the paper's equation labelled `eq:cubic-symm-enrich`*. Each shade of green denotes one three-qubit gate, not two successive two-qubit gates.*

At $a_{i,j,k}$, these are the two pairs listed above: the lower C-face with the far B-edge, and the near C-edge with the upper B-face. There are exactly two CCZ's per control in this pattern. This assignment is repeated at every vertex.

The paper denotes this local assignment by $\phi^{\mathsf{T}}(v)$. In this example $\phi^{\mathsf{T}}(v)$ is a set of **pairs** (e, e'), not merely a set of single qubits as in the SWAP example. In a basis-independent form, the enrichment circuit is

$$\Omega = \prod_{v} \prod_{(e,e')\in\phi^{\mathsf{T}}(v)} \mathsf{CCZ}_{a_v, B_e, C_{e'}}. \tag{179}$$

The map must be local and surjective in the paper's sense: every required physical CZ in U_{CZ} is assigned to an odd number of controls, preferably exactly one, and each control reaches only $O(1)$ nearby pairs. This is the CZ version of the sparse assignment $e \mapsto \phi(e)$ in the SWAP construction.

To see the symmetry enrichment explicitly, use the one-gate identity

$$X_a \,\mathsf{CCZ}_{a,B,C}\, X_a \,\mathsf{CCZ}_{a,B,C} = \mathsf{CZ}_{B,C}. \tag{180}$$

It follows just by checking $a = 0, 1$: flipping the control interchanges the "apply CZ" and "do nothing" branches. Multiply Eq. (180) over all temporary vertices and their assigned pairs. The odd-assignment condition then gives

$$X^a_{\text{all}} \Omega X^a_{\text{all}} \Omega^\dagger = U_{\mathsf{CZ}}. \tag{181}$$

This is the same group-commutator test used for graph-gauged SWAP, with controlled swaps replaced by controlled CZ gates. It is the one equation that shows the cubic ancilla is gauging the desired logical symmetry rather than merely adding extra qubits.

Gauge constraints, matter checks and commuting projectors

Measure every temporary a_v qubit in the X basis. For the moment, consider the all-+1 branch; the next subsection of this note explains how a decoder and feedforward correct the actual random signs. From Eq. (176), the temporary-vertex stabilizer becomes the surviving gauge constraint

$$\widetilde{K}_v^A = \left(\prod_{f\ni v} X_f^A\right)\left(\prod_{(e,e')\in\phi^{\mathsf{T}}(v)} \mathsf{CZ}_{B_e,C_{e'}}\right). \tag{182}$$

The first factor is an ordinary X-type gauge transformation on the six edges incident to a cubic vertex. The second factor simultaneously performs the local piece of the original B-C symmetry. This is the concrete form of the slogan "a gauge transformation carries the old symmetry."

The matter checks become dressed as well. The useful one-qubit identity is

$$\mathsf{CCZ}_{a,B,C} X_B \mathsf{CCZ}_{a,B,C} = X_B \mathsf{CZ}_{a,C}, \tag{183}$$

and the analogous equation holds with B and C exchanged. Therefore each X-type ALP check picks up controlled-Z factors involving temporary ancillas. Before measurement, those factors can be traded, using products of the edge relations K_f^A, for operators on paths of surviving A-edge qubits - the path-replacement mechanism developed in the graph-gauged SWAP example. The original ALP Z-checks themselves remain unchanged, since CCZ is diagonal and they act only on matter qubits. Products of K_f^A around a closed ancilla loop also survive: every temporary Z_v^a occurs twice, leaving only the loop product of Z_f^A. On the cubic lattice the elementary examples are the four-edge plaquette flux checks. The complicated dressing belongs to the matter X-checks and the combined gauge constraints; it should not obscure these simple unchanged Z-type operators.

This explains why the paper presents the final stabilizers mainly as 3D figures rather than as a short one-line formula. Three distinct incidence structures must be superimposed: the six-qubit ALP checks, the cubic ancilla graph, and the local B-C pair assignment. The final operators are local, but "local" now means local in that combined three-layer incidence structure, not local in a single planar projection.

There is a second subtlety. The raw dressed generators can commute only **up to** the gauged U_{CZ} symmetry. To define a Hamiltonian with mutually commuting projectors, the paper symmetrizes them with respect to that $\mathbb{Z}_2$ action. Commutation on the constrained codespace is weaker than commutation as operators on the entire Hilbert space; a commuting-projector Hamiltonian requires the latter. A bare matter-check label can be changed by the gauge transformation, so the gauge-invariant term must include both members of its symmetry orbit.

For a concrete illustration of an orbit, the transversal symmetry above sends A_v^B to $A_v^B B_c^C$ and back again, while fixing B_c^C. Here an *orbit* simply means the set of operators obtained by repeatedly applying the symmetry. The simultaneous positive-eigenvalue projector

$$Q = \frac{I + A_v^B}{2}\frac{I + B_c^C}{2}$$

is invariant under this exchange: replacing A_v^B by $A_v^B B_c^C$ leaves Q unchanged, because $B_c^C(I + B_c^C) = I + B_c^C$. The condition $B_c^C = +1$ makes the two representatives of the vertex check equivalent. This elementary example explains why imposing the relevant constraints can turn a symmetry orbit into one well-defined projector. For the gauged operators one must also include the gauge constraints; the paper's

appendix on graph-gauged ALP fractons carries out the local dressing calculation. Simply averaging two arbitrary operators is not, by itself, a proof that the result is a projector or that different terms commute.

From the ALP construction to logical measurement

The SWAP example taught the general graph procedure. This ALP example adds the specific ingredient needed later for computation:

$$\text{temporary control} \quad + \quad \mathsf{CCZ} \quad \Longrightarrow \quad \text{gauged transversal logical } \mathsf{CZ}. \tag{184}$$

The later ungauging protocol reverses the ancilla construction and measures the gauge qubits. Equation (181) then reappears as a projector onto an eigenvalue of the logical CZ acting on the original B, C blocks. On appropriate logical input states, that Clifford measurement prepares a non-stabilizer state. Thus the CCZ gates used here are not being claimed as directly applied logical non-Clifford gates; rather, they are the local measurement gadget from which a non-Clifford *resource state* emerges after gauging and ungauging.

The construction therefore connects four ingredients:

1. The ALP code has qubits on xy plaquettes and z edges, giving six-qubit cube and vertex checks.
2. Two copies possess a local transversal CZ matching.
3. A cubic graph cluster state plus local CCZ enrichment satisfies the global commutator identity (181).
4. Measuring the temporary ancillas turns this into a non-Abelian gauge constraint (182); ungauging later converts the same construction into a logical CZ measurement.

Returning to Section III.E: measurement preparation of non-Abelian qLDPC codes

Before using the ALP construction for computation, we return to the measurement-preparation argument of Section III.E. It supplies the missing operational step in the example: handling random measurement signs. The gauging discussion has used an idealized instruction: "project every temporary ancilla a_v onto $|+\rangle$." A physical X-basis measurement cannot promise that outcome. It returns a random number

$$x_v \in \{+1, -1\}, \qquad X_v^a \longmapsto x_v. \tag{185}$$

This section explains why that randomness does not turn gauging into a postselected protocol. For preparation from the symmetry-invariant input sector, and assuming that the Abelian code on the ancilla complex has a decoder, the random local signs can be removed deterministically by a Pauli correction on the surviving A qubits. The word *deterministic* means that every possible measurement record is accepted and corrected; it does not mean that individual measurement outcomes cease to be random.

First ignore the matter codes: measuring a graph cluster state

It is useful to strip the problem to its simplest form. Before the measurement, the temporary-vertex graph-cluster stabilizer is

$$K_v^a = X_v^a X_{d_0^a v}^A. \tag{186}$$

For an ordinary graph, $X^A_{d_0^a v}$ is the product of X on all A-edge qubits incident to vertex v. Since $K^a_v = +1$ on the cluster state, measuring $X^a_v = x_v$ leaves the surviving A qubits with the signed check

$$x_v X^A_{d_0^a v} = +1. \tag{187}$$

The usual Z-type ancilla checks, labelled schematically by $Z^A_{\partial_2[p]}$, are unaffected. Thus the only difference between the desired Abelian ancilla code and the actually prepared one is the list of signs $\{x_v\}$ in Eq. (187).

Turn these signs into a binary syndrome vector,

$$s = \sum_v s_v[v] \in A_0, \qquad s_v = \begin{cases} 0, & x_v = +1, \\ 1, & x_v = -1. \end{cases} \tag{188}$$

Why are there an even number of -1 outcomes in the clean target sector? Multiply every premeasurement vertex stabilizer in Eq. (186). Each A-edge occurs at the two ends of that edge, so its two factors cancel:

$$\prod_v K^a_v = \left(\prod_v X^a_v\right)\left(\prod_{f\in A_1} (X^A_f)^2\right) = \prod_v X^a_v = X^a_{\text{all}}. \tag{189}$$

All the K^a_v's have eigenvalue $+1$, hence the cluster state has $X^a_{\text{all}} = +1$. The individual X^a_v measurements commute, so their outcomes must obey

$$\prod_v x_v = +1. \tag{190}$$

A product of signs is $+1$ precisely when an even number of its factors are negative. For three temporary qubits, for example, $(+,+,+)$ and $(+,-,-)$ are allowed, whereas $(-,+,+)$ is not. After symmetry enrichment the corresponding global relation is $X^a_{\text{all}} U = +1$. In the clean target-preparation sector, the input matter state has $U = +1$, so the same even-parity conclusion follows. An odd parity would instead record the opposite global symmetry sector, not an ordinary local syndrome that an A-edge correction can erase.

This is exactly the familiar CSS situation: a -1 outcome says that the corresponding X-check has the wrong sign. On a 2D surface-code graph, one would draw the -1 vertices and connect them pairwise by a path of edges. In a general qLDPC complex there may be no literal picture, but the same binary linear-algebra question remains: find a 1-chain

$$r \in A_1 \quad \text{such that} \quad \partial^a_1 r = s. \tag{191}$$

For ideal measurements, finding any such chain suffices to correct these signs. In error correction with physical noise, a decoder uses a noise model to seek a likely error representative; choosing the logical class then also matters. It is not necessary for this discussion to know how that algorithm works internally.

The corresponding quantum correction is the Pauli product

$$Z_r^A := \prod_{f \in A_1} (Z_f^A)^{r(f)}. \tag{192}$$

Why does it fix the signs? A factor Z_f^A anticommutes with an X-check precisely when edge f occurs in that check. Therefore

$$Z_r^A X_{d_0^a v}^A Z_r^A = (-1)^{v(\partial_1^a r)} X_{d_0^a v}^A = (-1)^{s_v} X_{d_0^a v}^A. \tag{193}$$

Because $x_v = (-1)^{s_v}$, conjugating Eq. (187) by Z_r^A changes it to the desired $+X_{d_0^a v}^A = +1$ check. The correction commutes with all Z-type ancilla checks, since it consists only of Z's. This is the full deterministic-preparation argument for the Abelian ancilla code.

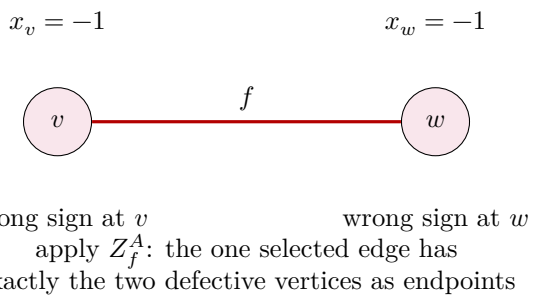
Worked nearest-neighbour example: two -1 outcomes.

Take two neighbouring vertices v and w of the cubic ancilla graph, joined by one surviving A-edge $f = \langle v, w \rangle$. Suppose that the two temporary measurements give $x_v = x_w = -1$, while every other temporary measurement gives $+1$. This is the smallest nontrivial syndrome compatible with a closed, periodic graph:

$$s = [v] + [w] = \partial_1^a [f]. \tag{194}$$

The decoder can therefore choose the shortest possible correction chain, $r = [f]$. Its quantum correction contains just **one** single-qubit Pauli:

$$Z_r^A = Z_f^A. \tag{195}$$



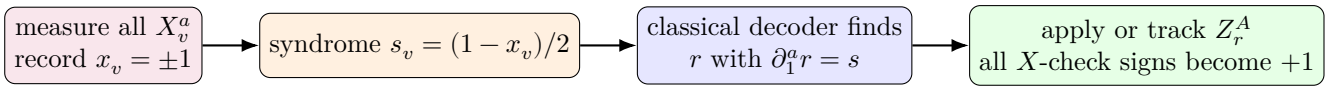
The two signed checks before correction are

$$-X_{d_0^a v}^A = +1, \qquad -X_{d_0^a w}^A = +1. \tag{196}$$

Edge f occurs once in each of the two vertex checks. Hence

$$Z_f^A X_{d_0^a v}^A Z_f^A = -X_{d_0^a v}^A, \qquad Z_f^A X_{d_0^a w}^A Z_f^A = -X_{d_0^a w}^A. \tag{197}$$

The correction flips both minus signs to plus signs at once. At every other vertex u, the edge f does not occur in $d_0^a u$, so the u-check is unchanged. In the gauged code, exactly the same calculation applies to $-X_{d_0^a v}^A \mathcal{U}_v$ and $-X_{d_0^a w}^A \mathcal{U}_w$, because Z_f^A commutes with the matter dressing $\mathcal{U}$. This is the most concrete example of the general rule $\partial_1^a r = s$.



Why exactly the same decoder works after gauging

Now restore the matter blocks B, C and the enrichment circuit. After the X_a measurements, the gauged vertex constraint has the general form

$$\mathcal{G}_v[x_v] = x_v X^A_{d_0^a v} \mathcal{U}_v. \tag{198}$$

The factor $\mathcal{U}_v$ is the local piece of the old transversal symmetry. In the SWAP example it is a product of SWAP_{B_e,C_e}'s; in the ALP CZ example it is a product of $\mathsf{CZ}_{B_e,C_{e'}}$'s. Crucially, the random outcome x_v occurs in exactly the same place as it did in Eq. (187): it merely multiplies the X^A factor by a sign.

The correction Z_r^A has two equally important properties:

1. By Eq. (193), it flips precisely the unwanted signs in $\mathcal{G}_v[x_v]$.
2. It commutes with $\mathcal{U}_v$, because Z_r^A acts on the surviving gauge qubits whereas $\mathcal{U}_v$ acts on B, C. It also commutes with the other Z-type and matter constraints.

Hence the same correction selected by the *Abelian* ancilla decoder turns

$$\mathcal{G}_v[x_v] \quad \longmapsto \quad \mathcal{G}_v[+1] \tag{199}$$

without changing the non-Abelian matter dressing. This is the central observation of the paper's subsection. Measurement randomness creates only an Abelian syndrome, even though the final code has non-Abelian structure.

What "feedforward" means here

The decoder may be a nonlocal classical computation: it reads the full list of outcomes before deciding on r. But once r is known, every factor of Z_r^A is a one-qubit Pauli. They can all be performed in one quantum layer. Often one does not physically apply them at all. One instead stores the binary vector r as a **Pauli frame** and flips the interpretation of later measurements accordingly.

This is why the paper calls the preparation a finite-depth *adaptive* circuit: the quantum circuit has bounded depth, while a classical decoder and a classically chosen Pauli frame connect the measurement step to later operations. This is a deterministic state-preparation statement assuming an appropriate decoder exists. It is not yet a complete proof of fault tolerance against noisy gates and noisy measurements; the paper identifies that stronger question as future work.

What to retain for the computation section

The ideal $|+\rangle_a$ projection used in a formal gauging derivation is therefore operationally replaced by

$$\text{measure } X_v^a \quad \longrightarrow \quad \text{decode the sign syndrome} \quad \longrightarrow \quad \text{update by } Z_r^A. \tag{200}$$

This prepares the desired $+1$ constraints in the invariant input sector. The next section asks a different question: what happens for an **arbitrary logical input**, possibly a superposition of symmetry sectors? Then the global products of outcomes contain information about that input and must be retained.

Indeed, an edge correction changes a sign syndrome by $\partial_1^a r$. For any cocycle μ, its effect on the corresponding outcome parity is zero because $\mu(\partial_1^a r) = (d_0^a \mu)(r) = 0$. Thus local feedforward cannot erase a logical symmetry outcome. This is why deterministic preparation and logical measurement are compatible, but are not the same protocol on the same class of inputs.

Section VI.A: logical Clifford measurement from gauging and ungauging

We now reach the computational claim of the paper. Gauging was introduced as a way to make a transversal symmetry local and dynamical. **Ungauging** reverses that enlargement: undo the ancilla cluster entangler and measure away the surviving A gauge qubits. The surprising result is that the combined operation is a measurement of the logical symmetry that was gauged.

The paper writes the derivation for a transversal CZ in the homological construction, where there can be several cocycle symmetries. The conclusion is broader. In the connected-graph construction there is only one independent cocycle, so the same derivation reduces to a measurement of its chosen involutory symmetry, such as SWAP or the CZ considered here.

The claim in one line

Let U_μ be the transversal logical Clifford symmetry with $U_\mu^\dagger = U_\mu$ and $U_\mu^2 = I$, associated with an ancilla 0-cocycle $\mu \in \ker d_0^a$. Gauging and then ungauging applies the projector

$$P_{x_\mu} = \frac{I + x_\mu U_\mu}{2} \tag{201}$$

to the two input code blocks. Here $x_\mu = \pm 1$ is obtained from the temporary-ancilla measurement outcomes. Equation (201) is precisely the projective measurement of the Hermitian involution U_μ: it selects its $+1$ or -1 eigenspace.

For the connected graph used in the SWAP and ALP examples, $\ker d_0^a$ is one-dimensional: it contains zero and the uniform nonzero cocycle μ_{all}. The formula is then simply

$$\frac{I + x\,U}{2}. \tag{202}$$

The rest of this section derives this statement. The logical information never has to be decoded into individual physical qubits; all operations are performed on the qLDPC blocks and the ancillary qubits.

Ingredients and notation

Let $|\Psi\rangle_{BC}$ be an arbitrary state in the two input code spaces. Let C_A be the Clifford circuit which prepares the ancilla cluster state from

$$|0\rangle_A^{\otimes} |+\rangle_a^{\otimes}. \tag{203}$$

In the Hadamard convention used here, C_A is a product of CNOT gates from temporary controls a_v to the adjacent A qubits. The complete measurement procedure is:

1. Apply C_A, then the enrichment circuit Ω.
2. Measure each a_v in the X basis and record x_v.
3. Apply $C_A^\dagger$, then measure each A_e in the Z basis, recording its bit $z(e)$.
4. Solve $d_0^a \phi_0 = z$, and apply $X_{\phi_0}^a$.

5. Apply $\Omega^\dagger$. The ancillary registers now factor from the matter state and can be discarded.

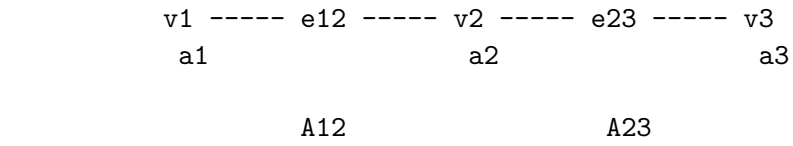
The measured a qubits must remain available for the later gates. An ideal nondestructive X measurement leaves them in the recorded X-eigenstates. With destructive readout, reprepare those same $|x_v\rangle_X$ states before step 3. Discarding them without this replacement would not implement the displayed protocol.

The correction in step 4 removes the offset ϕ_0, not the logical measurement outcome. It is required independently of whether x_{ϕ_0} is positive or negative: that scalar is only a global phase. A software frame can replace the physical correction only if its propagation through the subsequent enrichment inverse is accounted for.

We next recall the ancillary chain complex, then derive why these operations produce a projector on the original encoded state.

Chains and cochains on the ancillary register

The calculation uses two different maps on the ancillary register. A boundary locates the endpoints of a correction string; a coboundary tells us which gauge qubits are flipped by a choice of temporary controls. Keeping these roles separate is easiest on a three-vertex graph:



Each vertex v carries a temporary qubit a_v, and each edge e carries a surviving gauge qubit A_e. This is the same vertex-and-edge picture used in ordinary lattice gauging. The qLDPC construction permits the graph to be an abstract incidence structure rather than a planar drawing, but the following bookkeeping is identical.

The symbols A_0 and A_1 mean, respectively, the binary vector spaces of collections of vertices and collections of edges:

mathematical object	plain-English meaning	physical qubits
A_0	collections of vertices	a_v live at vertices
A_1	collections of edges	A_e live on edges.

Because the coefficients are in $\mathbb{F}_2$, a collection means "mark some objects," and marking an object twice cancels: $1 + 1 = 0 \pmod 2$. For example,

$$r = e_{12} + e_{23} \in A_1$$

means "mark both edges," while

$$s = v_1 + v_3 \in A_0$$

means "mark vertices v_1 and v_3." These are **chains**.

The boundary map turns a marked set of edges into the marked endpoints left over:

$$\partial_1^a(e_{12}) = v_1 + v_2, \qquad \partial_1^a(e_{23}) = v_2 + v_3,$$

and therefore

$$\partial_1^a(e_{12} + e_{23}) = (v_1 + v_2) + (v_2 + v_3) = v_1 + v_3.$$

The interior vertex v_2 cancels because the two selected edges meet there. This is the familiar fact that a path has endpoints but its interior has no boundary. In measurement decoding, the equation

$$\partial_1^a r = s$$

means: choose an edge string r whose endpoints are exactly the vertices with unwanted measurement signs s.

Now introduce a 0-cochain $\phi \in A^0$. For intuition, draw it as another assignment of bits to vertices:

```
vertex:       v1   v2   v3
phi(v):        1    0    1
```

The coboundary $d_0^a\phi$ assigns bits to edges. On an edge, it asks whether the endpoint bits differ:

$$(d_0^a\phi)(e_{12}) = \phi(v_1) + \phi(v_2) = 1 + 0 = 1,$$

$$(d_0^a\phi)(e_{23}) = \phi(v_2) + \phi(v_3) = 0 + 1 = 1.$$

Thus $d_0^a\phi$ marks the edges crossing from a vertex with bit 0 to one with bit 1:

```
vertex bits phi:       1 ---- 0 ---- 1
edge bits d phi:           1       1
```

The distinction between the two maps is therefore

$$\partial_1^a : \underbrace{\text{marked edges}}_{\text{chain}} \longrightarrow \underbrace{\text{their endpoints}}_{\text{chain}},$$

$$d_0^a : \underbrace{\text{vertex-bit assignment}}_{\text{cochain}} \longrightarrow \underbrace{\text{edge-bit assignment}}_{\text{cochain}}.$$

In the measurement-and-ungauging calculation, ϕ is used only to choose a term in an expansion. Its coboundary specifies the surviving edge qubits on which to apply X:

$$X^A_{d_0^a\phi} = \prod_{e:\,(d_0^a\phi)(e)=1} X_{A_e}.$$

For the displayed example, this gives

$$X^A_{d_0^a\phi} = X_{A_{12}} X_{A_{23}}, \qquad X^A_{d_0^a\phi}|0\rangle_A = |1\rangle_{A_{12}}|1\rangle_{A_{23}}.$$

Thus a subsequent Z_A measurement records two minus signs. Its edge-outcome pattern is written as the 1-cochain $z = d_0^a\phi$.

The precise definitions underlying these bit assignments are

$$A^0 = \mathrm{Hom}(A_0, \mathbb{F}_2), \qquad A^1 = \mathrm{Hom}(A_1, \mathbb{F}_2).$$

Formally, cochains are linear functions. But because there is a preferred basis of vertices and edges, such a function is completely specified by one bit per vertex or edge. That is why a cochain can be drawn just like a marked collection. For a first reading, retain the following rule:

```
chain:    "Which vertices or edges have I selected?"
cochain:  "What bit do I assign to each vertex or edge?"
```

They look alike in coordinates, but they participate in different maps:

$$\partial : \text{chains} \longrightarrow \text{chains}, \qquad d : \text{cochains} \longrightarrow \text{cochains}.$$

For this three-vertex example, the two maps are the matrices

$$\partial_1^a = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad d_0^a = (\partial_1^a)^{\mathsf{T}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Multiplying the first by the edge-chain column $(1,1)^{\mathsf{T}}$ gives the endpoint column $(1,0,1)^{\mathsf{T}}$. Multiplying the second by the vertex-cochain column $(1,0,1)^{\mathsf{T}}$ gives the edge bits $(1,1)^{\mathsf{T}}$. All matrix arithmetic is modulo two. The transpose relation follows from the definition

$$(d_0^a \phi)(r) = \phi(\partial_1^a r).$$

The left side evaluates an edge cochain on an edge chain; the right side evaluates the vertex cochain on that chain's boundary. For a single edge, this equation says to add the two endpoint bits. For a path, all interior bits cancel.

In homological gauging, A_0, A_1 still label temporary and surviving qubits, but the incidence matrix need not be that of an ordinary graph. An A qubit can then couple to a different set of temporary controls. The matrix equations, dual-space definitions and calculation below still apply; the statement "add the two endpoint bits" is specific to a graph. Consequently $\ker d_0^a$ can describe subsystem symmetries rather than just a uniform bit assignment.

Step 1: expand the fixed X_a-measurement record

For a record $x_v = \pm 1$, the one-qubit measurement projector is

$$\Pi_v(x_v) = \frac{I + x_v X_v^a}{2}. \tag{204}$$

The cluster relation lets us move this projector through the preparation circuit. In the notation used throughout the note,

$$\Pi_x \, \Omega C_A(|\Psi\rangle_{BC}|0\rangle_A|+\rangle_a) \;=\; C_A \prod_v \frac{I + x_v X_v^a X_{d_0^a v}^A}{2} \Omega(|\Psi\rangle_{BC}|0\rangle_A|+\rangle_a). \tag{205}$$

This equation says: "instead of viewing the X_a measurements as acting after cluster preparation, we can pull their projectors through the cluster-preparation circuit." In that new viewpoint, each measured temporary qubit is accompanied by X's on all adjacent surviving A-qubits.

Write the initial state as

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|0\rangle_A|+\rangle_a.$$

For a particular set of recorded measurement outcomes $x = \{x_v\}$, with $x_v = \pm 1$,

$$\Pi_x = \prod_v \Pi_v(x_v), \qquad \Pi_v(x_v) = \frac{I + x_v X_v^a}{2}.$$

This is just the projector onto the stated outcomes: if $x_v = +1$, it projects a_v onto $|+\rangle$; if $x_v = -1$, onto $|-\rangle$.

The cluster-preparation circuit C_A is a product of CNOTs from a temporary vertex qubit a_v to each neighbouring gauge-field qubit A_e. Its basic conjugation rule is

$$C_A^\dagger X_v^a C_A = X_v^a X_{d_0^a \mathbf{v}}^A.$$

Here $d_0^a\mathbf{v}$ is the binary list of edges incident on v. Thus, if v has neighbouring edges $e_1, \ldots, e_r$,

$$X_{d_0^a \mathbf{v}}^A = X_{A_{e_1}} \cdots X_{A_{e_r}}.$$

For example, for a degree-three vertex,

$$C_A^\dagger X_v^a C_A = X_{a_v} X_{A_{e_1}} X_{A_{e_2}} X_{A_{e_3}}.$$

Now insert $I = C_A C_A^\dagger$ around each measurement projector:

$$\Pi_v(x_v) C_A = C_A (C_A^\dagger \Pi_v(x_v) C_A) = C_A \frac{I + x_v X_v^a X_{d_0^a \mathbf{v}}^A}{2}.$$

Doing this for every vertex gives the displayed equation:

$$\Pi_x \Omega C_A |\Phi_0\rangle = C_A \prod_v \frac{I + x_v X_v^a X_{d_0^a \mathbf{v}}^A}{2} \Omega |\Phi_0\rangle.$$

The enrichment circuit Ω commutes with C_A: both are controlled by the a-qubits, but they act on different targets (Ω on B, C, C_A on A). That is why their order can be exchanged here.

So the important content is not a new operation. It is a change of description:

measuring X_{a_v} after cluster preparation	$\longleftrightarrow$	measuring the cluster stabilizer $X_{a_v} X_{d_0^a \mathbf{v}}^A$ before it.

These are exact identities for unnormalized post-measurement states; the factors of $1/2$ have been retained. The crucial feature is the factor $X_v^a X_{d_0^a v}^A$: measuring an a qubit is tied to an X operation on the adjacent A-gauge qubits.

Expanding the product in Eq. (205) means choosing either the I term or the $X_v^a X_{d_0^a v}^A$ term at every vertex. A choice is a binary 0-cochain $\phi \in A^0$. Concretely, this means that ϕ is a binary label assigned to every temporary-ancilla vertex:

$$\phi : V_A \longrightarrow \mathbb{F}_2 = \{0, 1\}.$$

If there are N temporary a-qubits, ϕ is simply an N-bit string. In this equation it is not a new quantum state, a measurement outcome, or a physical excitation. It is an accounting device for one term in the algebraic expansion. At each vertex v, it records which of the two terms was selected:

$\phi(v)$	selected term
0	I
1	$x_v X_v^a X_{d_0^a v}^A$.

For example, for three vertices, $\phi = (1, 0, 1)$ means: select the non-identity term at vertices 1 and 3, and select I at vertex 2. The corresponding contribution is $x_1 x_3 K_1 K_3$, where $K_v = X_v^a X_{d_0^a v}^A$. Therefore

$$\prod_v \left(I + x_v X_v^a X_{d_0^a v}^A\right) = \sum_{\phi \in A^0} x_\phi \, X_\phi^a X_{d_0^a \phi}^A, \qquad x_\phi := \prod_v x_v^{\phi(v)}. \tag{206}$$

The subscript on x_ϕ denotes a product of signs, whereas the entries of ϕ are exponents:

$$x_\phi := \prod_v x_v^{\phi(v)},$$

not $\prod_v x_{\phi(v)}$. The exponent is essential. Since $\phi(v)$ is either 0 or 1,

$$x_v^{\phi(v)} = \begin{cases} 1, & \phi(v) = 0, \\ x_v, & \phi(v) = 1. \end{cases}$$

Thus x_ϕ is simply the product of the recorded signs x_v over those vertices selected by ϕ.

This is the ordinary binomial expansion, with $K_v = X_v^a X_{d_0^a \mathbf{v}}^A$ and $X_\phi^a = \prod_v (X_v^a)^{\phi(v)}$. The two-vertex case displays every term explicitly.

For two vertices v_1, v_2, the left-hand side is

$$(I + x_1 K_1)(I + x_2 K_2) = I + x_1 K_1 + x_2 K_2 + x_1 x_2 K_1 K_2.$$

The four terms correspond exactly to the four binary cochains:

$\phi = (\phi(v_1), \phi(v_2))$	x_ϕ	Pauli operator
$(0,0)$	1	I
$(1,0)$	x_1	K_1
$(0,1)$	x_2	K_2
$(1,1)$	$x_1 x_2$	$K_1 K_2$.

Finally, the compact form of the Pauli operator follows because d_0^a is linear over $\mathbb{F}_2$:

$$d_0^a \phi = \sum_v \phi(v)\, d_0^a \mathbf{v}.$$

Therefore, selecting K_v at every vertex with $\phi(v) = 1$ gives

$$\prod_{v:\phi(v)=1} K_v = \left[\prod_v (X_v^a)^{\phi(v)}\right] \left[\prod_v \left(X^A_{d_0^a \mathbf{v}}\right)^{\phi(v)}\right] = X^a_\phi X^A_{d_0^a \phi}.$$

The missing factors of $1/2$ are intentional: Eq. (206) expands the numerators of the measurement projectors. The full projector has an overall factor $2^{-|V_A|}$, which is irrelevant once the state is normalized.

This is not a new physical superposition of measurement records. It is an algebraic expansion of the fixed-record projector. It is useful because the co-boundary $d_0^a \phi$ will be tested by the later Z_A measurements.

Step 2: the Z_A record fixes a coset of cocycles

Ungauging first applies $C_A^\dagger$, removing the leftmost cluster entangler in Step 1. The A register is not yet reset to zero: each summand contains $X^A_{d_0^a \phi}|0\rangle_A = |d_0^a \phi\rangle_A$. Let the subsequent Z_A measurement record be a binary 1-cochain z: $z(f) = 1$ means the outcome on edge f is -1.

Here the cleanest way to read the notation is: choose the natural bases of vertices and edges, then represent each cochain by its list of values on those basis objects. Thus $\phi \in A^0$ is formally a linear function on vertex chains, but we draw it as vertex bits; likewise, $z \in A^1$ is formally a linear function on edge chains, but we draw it as edge bits. The important point is that z and $d_0^a \phi$ belong to the same space, A^1. Equation (207) is an equality of two edge cochains, meaning

$$z(e) = (d_0^a \phi_0)(e) \qquad \text{for every edge } e.$$

Take the concrete vertex assignment

```
vertex:       v1   v2   v3
phi(v):        1    0    1
```

Its coboundary is

$$(d_0^a \phi)(e_{12}) = 1 + 0 = 1, \qquad (d_0^a \phi)(e_{23}) = 0 + 1 = 1.$$

Thus its edge values are

```
edge:             e12  e23
(d phi)(e):         1    1
```

The operator $X^A_{d_0^a\phi}$ applies X precisely on those edge qubits with assigned bit 1. Therefore

$$X^A_{d_0^a\phi}|0\rangle_A = X_{A_{12}}X_{A_{23}}(|0\rangle_{A_{12}}|0\rangle_{A_{23}}) = |1\rangle_{A_{12}}|1\rangle_{A_{23}}.$$

The later Z_A measurement reads this computational-basis string. With the convention $z(e) = 0$ for a $+1$ Z_{A_e} outcome and $z(e) = 1$ for a -1 outcome, the displayed state has only one possible record:

```
edge:       e12  e23
z(e):         1    1
```

Hence $z = d_0^a\phi$. If instead the measurement reported $z = (0, 1)$, its overlap with this branch would vanish:

$$\langle 0, 1|1, 1\rangle = 0.$$

This is the meaning of "its overlap with the observed record is nonzero only if" there is some 0-cochain ϕ_0 with

$$z = d_0^a\phi_0 \tag{207}$$

The subscript 0 says only: choose one cochain whose coboundary matches the actual record. It need not be unique. This is an outcome constraint, not a postselection condition: the actual record automatically has this form because it came from the state in Eq. (206).

Indeed, if $\mu \in \ker d_0^a$, then

$$d_0^a(\phi_0 + \mu) = d_0^a\phi_0 = z.$$

These alternative ϕ's are precisely the branches that remain after the Z_A measurement, and they differ by ancilla symmetries.

Now compare any other surviving summand ϕ with ϕ_0. It must obey

$$d_0^a\phi = d_0^a\phi_0 \quad \Longleftrightarrow \quad d_0^a(\phi + \phi_0) = 0. \tag{208}$$

Thus every surviving label has the form

$$\phi = \phi_0 + \mu, \qquad \mu \in \ker d_0^a. \tag{209}$$

The surviving set is the **coset** $\phi_0 + \ker d_0^a$: the set obtained by adding ϕ_0 to every vector in the kernel. Unless the observed edge string is zero, the surviving ϕ's are not themselves cocycles. Their differences are. Subtracting the known offset (the same as adding it over $\mathbb{F}_2$) leaves only the symmetry labels.

On the three-vertex line, take $z = (1, 1)$. The equations are $\phi_1 + \phi_2 = 1$ and $\phi_2 + \phi_3 = 1$, with exactly two solutions: $(1, 0, 1)$ and $(0, 1, 0)$. Their difference is $(1, 1, 1)$, which has zero coboundary. Neither solution has zero coboundary. This small example is why the coset, rather than just the kernel, appears before feedforward.

The binary string z is a **coboundary**, belonging to $\operatorname{im} d_0^a$. On a graph with loops, not every edge string has this property: around any loop the endpoint differences sum to zero. These restrictions hold automatically for ideal outcomes in this protocol; no experiment is rejected to enforce them.

The offset ϕ_0 is an ordinary measurement byproduct. Applying $X^a_{\phi_0}$ before undoing enrichment removes it. After this feedforward, the part that acts nontrivially on the input blocks is

$$\sum_{\mu \in \ker d_0^a} x_\mu X_\mu^a \, \Omega. \tag{210}$$

Notice the contrast with the previous measurement-preparation section. Local byproducts such as ϕ_0 are correctable. The signs x_μ attached to genuine cocycles are not local syndrome data to erase: they are the logical measurement outcomes we want to retain.

Step 3: remove the ancilla and reveal the logical projector

Undo the enrichment circuit and use its defining symmetry-enrichment identity,

$$\Omega^\dagger X_\mu^a \Omega = X_\mu^a U_\mu. \tag{211}$$

After conjugating through Ω, the remaining ancilla operators act on the original $|+\rangle_a$ factors. Hence each X_μ^a acts as $+1$, and the resulting state factors into an ancilla product state and a matter state. Choose a basis $\{\mu_i\}$ of the cocycle space. The output on the input codes is, up to normalization,

$$|\Psi\rangle_{BC} \longmapsto \prod_i \frac{I + x_{\mu_i} U_{\mu_i}}{2} |\Psi\rangle_{BC}. \tag{212}$$

What $\mu \in \ker d_0^a$ means here The kernel belongs to the map d_0^a: it is the vector space of inputs that this map sends to zero. Saying that μ belongs to it means

$$\mu \in \ker d_0^a \qquad \Longleftrightarrow \qquad d_0^a \mu = 0.$$

For an ordinary ancilla graph, on an edge $e = \langle v, w \rangle$, this condition says

$$(d_0^a \mu)(e) = \mu(v) + \mu(w) = 0 \pmod 2.$$

Thus neighbouring vertices must have equal μ-bits. On a connected graph, μ must be constant everywhere, leaving just two possibilities:

```
mu = 0:       0 ---- 0 ---- 0 ---- ...
mu = mu_all:  1 ---- 1 ---- 1 ---- ...
```

For the three-vertex line used above,

$$d_0^a(m_1, m_2, m_3) = (m_1 + m_2,\ m_2 + m_3),$$

so $d_0^a \mu = 0$ requires $m_1 = m_2 = m_3$. Hence

$$\ker d_0^a = \{(0,0,0),(1,1,1)\}.$$

Why do precisely these cochains survive Step 2? Suppose one vertex cochain ϕ_0 explains the final edge record:

$$z = d_0^a \phi_0.$$

For any $\mu \in \ker d_0^a$,

$$d_0^a(\phi_0 + \mu) = d_0^a \phi_0 + d_0^a \mu = z.$$

Therefore ϕ_0 and $\phi_0 + \mu$ create exactly the same bit pattern on every surviving A-edge qubit. The final Z_A measurements cannot distinguish them. In a connected graph, adding μ_{all} simply flips every vertex bit; each edge still has the same endpoint *difference*. The kernel is therefore the residual global ambiguity left after the edge measurements have removed all local information.

Step 3 translates that ambiguity into a logical operation. The enrichment circuit was chosen so that every such ancilla symmetry obeys

$$\Omega^\dagger X_\mu^a \Omega = X_\mu^a U_\mu.$$

Moving X_μ^a backward through enrichment produces the same ancilla operator times the logical Clifford symmetry U_μ on the B, C code blocks. The a qubits are in $|+\rangle_a$, so X_μ^a acts as $+1$ and disappears. What remains is

$$\sum_{\mu \in \ker d_0^a} x_\mu U_\mu.$$

For a connected graph, the kernel has only 0 and μ_{all}. Dividing this operator sum by two gives the projector

$$\frac{I + x_{\text{all}} U_{\text{all}}}{2}, \qquad x_{\text{all}} = x_{\mu_{\text{all}}} = \prod_v x_v.$$

Although every temporary qubit was measured separately, only this global parity of their outcomes remains after ungauging. It is exactly the outcome of measuring the logical symmetry U_{all}.

This is Eq. (201), now derived. In the homological construction, each independent cocycle can measure a related, potentially addressable logical CZ. In the connected-graph construction, there is one independent cocycle and Eq. (212) is one logical-gate measurement.

Why a sum over cocycles becomes a product of projectors Let $k = \dim \ker d_0^a$. Choosing a basis means that every cocycle has one unique expression $\mu = \sum_{i=1}^k t_i \mu_i$ with bits t_i. In this CZ construction the symmetry operators are commuting involutions and obey $U_{\mu+\nu} = U_\mu U_\nu$; the outcome signs obey the same multiplication rule, since $x_v^2 = 1$. Therefore

$$\sum_{\mu\in\ker d_0^a} x_\mu U_\mu = \prod_{i=1}^{k}(I + x_{\mu_i} U_{\mu_i}).$$

For two basis cocycles this reads

$$I + x_{\mu_1}U_{\mu_1} + x_{\mu_2}U_{\mu_2} + x_{\mu_1}x_{\mu_2}U_{\mu_1}U_{\mu_2} = (I + x_{\mu_1}U_{\mu_1})(I + x_{\mu_2}U_{\mu_2}).$$

It is the same binomial expansion as Step 1, now restricted to the undistinguished symmetry labels. Each factor divided by two is a projector:

$$\left(\frac{I+xU}{2}\right)^2 = \frac{I + 2xU + U^2}{4} = \frac{I+xU}{2}, \qquad U\frac{I+xU}{2} = x\frac{I+xU}{2}.$$

The first identity says repeating the projection has no further effect; the second says its surviving states have eigenvalue x. Opposite-sign projectors are orthogonal, and their sum is I. The simultaneous projector is their product because these symmetries commute. Independent cocycles need not give independent logical gates: a symmetry acting as identity on the codespace has a fixed positive outcome.

For a normalized input, the probability of a logical record is $\|P_x|\Psi\rangle\|^2$, and the conditional output is $P_x|\Psi\rangle/\|P_x|\Psi\rangle\|$. The projector's factor of two and the state's normalization are different things. A general Clifford gate need not satisfy $U^2 = I$; the binary projector formula here applies to the involutions in this construction.

The complete protocol with the ancillary registers retained

The following calculation keeps the normalization factors and the measured A register, so that every equality has the same input and output spaces. Set $N = \dim A^0$, $k = \dim\ker d_0^a$, and define

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|0\rangle_A|+\rangle_a, \qquad P_z^A = \prod_e \frac{I + (-1)^{z(e)}Z_{A_e}}{2}.$$

For the fixed records x, z, the unnormalized state after undoing the cluster entangler and measuring A is

$$\begin{aligned} P_z^A C_A^\dagger \Pi_x \Omega C_A|\Phi_0\rangle &= 2^{-N}\sum_{\phi:\, d_0^a\phi = z} x_\phi X_\phi^a \Omega(|\Psi\rangle_{BC}|z\rangle_A|+\rangle_a) \\ &= 2^{-N}x_{\phi_0}X_{\phi_0}^a \sum_{\mu\in\ker d_0^a} x_\mu X_\mu^a \Omega(|\Psi\rangle_{BC}|z\rangle_A|+\rangle_a). \end{aligned}$$

The first line is Step 1's expansion followed by Step 2's selection rule. The second substitutes $\phi = \phi_0 + \mu$. Applying the correction $X_{\phi_0}^a$ and then $\Omega^\dagger$ gives exactly

$$2^{k-N}x_{\phi_0}\left[\prod_{i=1}^{k}\frac{I + x_{\mu_i}U_{\mu_i}}{2}\right]|\Psi\rangle_{BC}\otimes|z\rangle_A\otimes|+\rangle_a.$$

The factor x_{ϕ_0} is a global sign. The scalar 2^{k-N} contributes to the probability of the *full* ancillary record; it does not change the normalized conditional matter state. The A qubits are left in the measured string

$|z\rangle$, not automatically $|0\rangle$; they can be discarded or reset by X_z^A. The a qubits have returned to $|+\rangle$ and can also be discarded.

The physical roles of the three steps are now explicit. The temporary measurements introduce the sum over cochains. The gauge-qubit measurements identify their coboundaries, leaving a coset of the kernel. Correcting its offset and undoing enrichment converts the remaining symmetry sum into logical projectors. In a connected graph, the only logical measurement bit left is $x_{\text{all}} = \prod_v x_v$.

The magic-state example

Suppose the measured symmetry acts as one logical $\overline{\mathsf{CZ}}$ on a chosen pair of encoded qubits. This is a simple logical example of the preceding result, not an assertion that the whole-block ALP gate acts on only one pair: each ALP block encodes many qubits. If a logical gate acts on several pairs, one must analyze that full action, use an addressable construction, or fix suitable spectator logical qubits so the action reduces to the stated pair.

For that pair, choose the input logical state $|\overline{+}\rangle_B|\overline{+}\rangle_C$. A $+1$ logical CZ outcome gives

$$\begin{aligned} \frac{I+\overline{\mathsf{CZ}}}{2}|\overline{+}\rangle_B|\overline{+}\rangle_C &\propto |\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle, \\ |\overline{M}_{\mathsf{CZ}}\rangle &:= \frac{|\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle}{\sqrt{3}}. \end{aligned} \tag{213}$$

The $|\overline{11}\rangle$ component is absent because it is the unique computational-basis component with CZ eigenvalue -1. The state in Eq. (213) is non-stabilizer: a stabilizer state has a computational-basis support of size a power of two when its nonzero amplitudes have equal magnitude, whereas this state has three equal nonzero amplitudes. It is therefore a magic-state resource.

For this particular input, the $+1$ outcome occurs with probability $3/4$; the -1 outcome gives $|\overline{11}\rangle$. Thus this simple demonstration is heralded rather than a deterministic magic-state factory: retain a $+1$ trial or repeat the preparation. The paper's central achievement is that the required logical CZ measurement has been compiled into gauging, local measurements, decoding, and feedforward on qLDPC code blocks.

Applying a Clifford **unitary** to a stabilizer state would leave a stabilizer state. Measuring its eigenspaces is a different operation: $(I + \mathsf{CZ})/2$ deletes one of the four amplitudes rather than merely changing its sign. This is not a Pauli measurement. The physical enrichment uses non-Clifford CCZ gates, so the protocol does not obtain magic from Clifford gates and Pauli measurements alone. It converts that physical resource into an encoded non-stabilizer state.

From gauging to an encoded resource state

The construction is not using a transversal CCZ as a direct logical non-Clifford gate. It uses local controlled-symmetry enrichment, then removes the ancillary gauge system, to perform a **logical Clifford measurement**. On a suitable logical input, that measurement supplies the non-Clifford resource needed for universal computation.

Sources and scope

- Target text: arXiv-2602.12228v1/main.tex, Sections II.A–III.C, "Two Approaches to Gauging qLDPC Codes," and the introduction, "Graph requirements," and "Example: Gauging transversal

SWAP gate" of "Non-Abelian Codes from Gauging with Graph States," plus the opening graph-gauged ALP fracton example and "Measurement Preparation of Non-Abelian qLDPC Codes," and "Logical Clifford Measurement from Gauging and Ungauging" (Section VI.A).

- ALP illustrations: the original TikZ diagrams from source labels `eq:qubit-ALP-stabilizers`, `eq:cubic-CZ-sym`, and `eq:cubic-symm-enrich`, rendered with their original geometry and colours. Reproducible extracts are in figures/alp-paper.
- Context for gauging and $\mathcal{D}(D_4)$: arXiv-2209.06202v2/main.tex, especially its discussion of Abelian Kramers-Wannier gauging and D_4 quantum doubles.
- Experimental and twisted-$\mathbb{Z}_2^3$ presentation: arXiv-2305.03766v2/main.tex, especially "The model," "Ground State Preparation," and its table equating $\mathcal{D}(D_4)$ with $\mathcal{D}^\alpha(\mathbb{Z}_2^3)$.